

National Coverage Determination (NCD)

Positron Emission Tomography (FDG) for Oncologic Conditions

220.6.17

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Tracking Information

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Manual Section Title

Positron Emission Tomography (FDG) for Oncologic Conditions

Version Number**Effective Date of this Version**

11/10/2009

Ending Effective Date of this Version

08/04/2010

Implementation Date

01/04/2010

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Indications and Limitations of Coverage**General**

The Centers for Medicare and Medicaid Services (CMS) was asked to reconsider section 220.6, of the National Coverage Determinations (NCD) Manual, to end the prospective data collection requirements across all oncologic indications of FDG PET except for monitoring response to treatment. Section 220.6 of the NCD Manual, establishes the requirement for prospective data collection for FDG PET used in the diagnosis, staging, restaging, and monitoring response to treatment for brain, cervical, ovarian, pancreatic, small cell lung and testicular cancers, as well as for cancer indications not previously specified in section 220.6 in its entirety.

The CMS received public input indicating that the current coverage framework, which required cancer-by-cancer consideration of diagnosis, staging, restaging, and monitoring response to treatment should be replaced by a more omnibus consideration. Thus, CMS broadened the scope of this review through an announcement on the Web site and solicited additional public comment on the use of FDG PET imaging for solid tumors so that it could transparently consider this possibility.

1. Framework

Effective for claims with dates of service on and after April 3, 2009, CMS is adopting a coverage framework that replaces the four-part diagnosis, staging, restaging and monitoring response to treatment categories with a two-part framework that differentiates FDG PET imaging used to inform the initial antitumor treatment strategy from other uses related to guiding subsequent antitumor treatment strategies after the completion of initial treatment. CMS is making this change for all NCDs that address coverage of FDG PET for the specific oncologic conditions addressed in this decision.

2. Initial Anti-tumor Treatment Strategy

Effective for claims with dates of service on and after April 3, 2009, CMS has determined that the evidence is adequate to determine that the results of FDG PET imaging are useful in determining the appropriate initial treatment strategy for beneficiaries with suspected solid tumors and myeloma and improve health outcomes and thus are reasonable and necessary under §1862(a)(1)(A) of the Social Security Act (the Act).

Therefore, CMS will cover only one FDG PET study for beneficiaries who have solid tumors that are biopsy proven or strongly suspected based on other diagnostic testing when the beneficiary's treating physician determines that the FDG PET study is needed to determine the location and/or extent of the tumor for the following therapeutic purposes related to the initial treatment strategy:

- To determine whether or not the beneficiary is an appropriate candidate for an invasive diagnostic or therapeutic procedure; or
- To determine the optimal anatomic location for an invasive procedure; or

- To determine the anatomic extent of tumor when the recommended anti-tumor treatment reasonably depends on the extent of the tumor.

As exceptions to the April 3, 2009, initial treatment strategy section above:

- a. The CMS has reviewed evidence on the use of FDG PET imaging to determine initial anti-tumor treatment in patients with adenocarcinoma of the prostate. CMS has determined that the available evidence does not demonstrate that FDG PET imaging improves physician decision making in the determination of initial anti-tumor treatment strategy in Medicare beneficiaries who have adenocarcinoma of the prostate, does not improve health outcomes and is thus not reasonable and necessary under §1862(a)(1)(A) of the Act. Therefore, FDG PET is nationally non-covered for this indication of this tumor type.
- b. The CMS received no new evidence demonstrating a change was warranted with respect to the use of FDG PET imaging to determine initial anti-tumor treatment in breast cancer; thus CMS is not making any change to the current coverage policy for FDG PET in breast cancer. CMS is continuing to cover FDG PET imaging for the initial treatment strategy for male and female breast cancer only when used in staging distant metastasis. FDG PET imaging for diagnosis and initial staging of axillary nodes will remain non-covered.
- c. The CMS received no new evidence demonstrating a change was warranted with respect to use of FDG PET imaging of regional lymph nodes in melanoma; thus CMS is not changing the current NCD for FDG PET in melanoma. CMS will continue non-coverage of FDG PET for the evaluation of regional lymph nodes in melanoma. Other uses to determine initial treatment strategy remain covered.
- d. The CMS received no new evidence demonstrating a change was warranted with respect to use of FDG PET imaging in the initial treatment strategy for cervical cancer. CMS is continuing to cover FDG PET imaging as an adjunct test for the detection of pre-treatment metastasis (i.e., staging) in newly diagnosed cervical cancers following conventional imaging that is negative for extra-pelvic metastasis. All other uses of FDG PET for the initial treatment strategy for beneficiaries diagnosed with cervical cancer will continue to only be covered as research under §1862(a)(1)(E) of the Act through Coverage with Evidence Development (CED). Therefore, CMS will cover one initial FDG PET study for newly diagnosed cervical cancer when not used as an adjunct test for the detection of pre-treatment metastases following conventional imaging that is negative for extra-pelvic metastasis only when the beneficiary's treating physician determines that the FDG PET study is needed to inform the initial anti-tumor treatment strategy and the beneficiary is enrolled in, and the FDG PET provider is participating in, the

specific type of prospective clinical study outlined under subsequent treatment strategy below.

- e. Effective November 10, 2009, as a result of a reconsideration request, CMS ended the prospective data collection requirements, or coverage with evidence development, for the use of FDG PET imaging in the initial staging of cervical cancer related to initial treatment strategy. CMS is continuing to cover FDG PET imaging as an adjunct test for the detection of pre-treatment metastasis (i.e., staging) in newly diagnosed cervical cancers following conventional imaging that is negative for extra-pelvic metastasis.

Therefore, CMS will cover one initial FDG PET study for staging in beneficiaries who have biopsy-proven cervical cancer when the beneficiary's treating physician determines that the FDG PET study is needed to determine the location and/or extent of the tumor for the following therapeutic purposes related to initial treatment strategy:

- To determine whether or not the beneficiary is an appropriate candidate for an invasive diagnostic or therapeutic procedure; or
- To determine the optimal anatomic location for an invasive procedure; or
- To determine the anatomic extent of the tumor when the recommended anti-tumor treatment reasonably depends on the extent of the tumor.

Additionally, effective November 10, 2009, following a reconsideration request, CMS determines that there is no credible evidence that the results of FDG PET imaging are useful to make the initial diagnoses of cervical cancer, does not improve health outcomes, and is not reasonable and necessary under section 1862(a)(1)(A) of the Social Security Act. Therefore, CMS will non-cover FDG PET imaging for initial diagnosis of cervical cancer related to initial treatment strategy.

3. Subsequent Anti-tumor Treatment Strategy

As part of its April 3, 2009, NCD, the CMS reviewed evidence on the use of FDG PET in the subsequent treatment strategy for patients with tumor types other than those seven indications currently covered without exception (breast, colorectal, esophagus, head and neck (non-CNS/thyroid), lymphoma, melanoma, and non-small cell lung).

As a result, CMS determined that the available evidence is adequate to determine that FDG PET imaging also improves physician decision making in the determination of subsequent treatment strategy in Medicare beneficiaries who have ovarian cancer, cervical cancer, and myeloma, improves health outcomes, and is thus reasonable and necessary under §1862(a)(1)(A) of the Act.

Therefore, effective for claims with dates of service on and after April 3, 2009, for tumor types other than breast, colorectal, esophagus, head and neck (non-CNS/thyroid),

lymphoma, melanoma, non-small cell lung, ovarian, cervical, and myeloma, CMS has determined that the available evidence is not adequate to determine that FDG PET imaging improves physician decision making in the determination of subsequent anti-tumor treatment strategy or improves health outcomes in Medicare beneficiaries and thus is not reasonable and necessary under §1862(a)(1)(A) of the Act.

However, CMS has determined that the available evidence is sufficient to determine that FDG PET imaging for subsequent anti-tumor treatment strategy for all other tumor types other than the 10 indications noted above may be covered as research under §1862(a)(1)(E) of the Act through CED. Therefore, CMS will cover a subsequent FDG PET study for all other tumor types other than the 10 indications noted above, when the beneficiary's treating physician determines that the FDG PET study is needed to inform the subsequent anti-tumor treatment strategy and the beneficiary is enrolled in, and the FDG PET provider is participating in, the following type of prospective clinical study:

An FDG PET clinical study that is designed to collect additional information at the time of the scan to assist in patient management. Qualifying clinical studies must ensure that specific hypotheses are addressed; appropriate data elements are collected; hospitals and providers are qualified to provide the FDG PET scan and interpret the results; participating hospitals and providers accurately report data on all enrolled patients not included in other qualifying trials through adequate auditing mechanisms; and all patient confidentiality, privacy, and other Federal laws must be followed.

The clinical studies for which CMS will provide coverage must answer one or more of the following three questions:

Prospectively, in Medicare beneficiaries whose treating physician determines that the FDG PET study is needed to inform the subsequent anti-tumor treatment strategy, does the addition of FDG PET imaging lead to:

- A change in the likelihood of appropriate referrals for palliative care;
- Improved quality of life; or,
- Improved survival?

The study must adhere to the following standards of scientific integrity and relevance to the Medicare population:

- a. The principal purpose of the research study is to test whether a particular intervention potentially improves the participants' health outcomes.
- b. The research study is well-supported by available scientific and medical information or it is intended to clarify or establish the health outcomes of interventions already in common clinical use.

- c. The research study does not unjustifiably duplicate existing studies.
- d. The research study design is appropriate to answer the research question being asked in the study.
- e. The research study is sponsored by an organization or individual capable of executing the proposed study successfully.
- f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found in the Code of Federal Regulations (CFR) at 45 CFR Part 46. If a study is regulated by the Food and Drug Administration (FDA), it also must be in compliance with 21 CFR Parts 50 and 56.
- g. All aspects of the research study are conducted according to the appropriate standards of scientific integrity.
- h. The research study has a written protocol that clearly addresses, or incorporates by reference, the Medicare standards.
- i. The clinical research study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Trials of all medical technologies measuring therapeutic outcomes as one of the objectives meet this standard only if the disease or condition being studied is life-threatening as defined in 21 CFR 312.81(a) and the patient has no other viable treatment options.
- j. The clinical research study is registered on the www.ClinicalTrials.gov Web site by the principal sponsor/investigator prior to the enrollment of the first study subject.
- k. The research study protocol specifies the method and timing of public release of all pre-specified outcomes to be measured including release of outcomes if outcomes are negative or study is terminated early. The results must be made public within 24 months of the end of data collection. If a report is planned to be published in a peer-reviewed journal, then that initial release may be an abstract that meets the requirements of the International Committee of Medical Journal Editors. However, a full report of the outcomes must be made public no later than 3 years after the end of data collection.
- l. The research study protocol must explicitly discuss subpopulations affected by the treatment under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria affect enrollment of these populations, and a plan for the retention and reporting of said populations on the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary.

- m. The research study protocol explicitly discusses how the results are or are not expected to be generalizable to the Medicare population to infer whether Medicare patients may benefit from the intervention. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

Consistent with section 1142 of the Act, the Agency for Healthcare Research and Quality (AHRQ) supports clinical research studies that CMS determines meet the above-listed standards and address the above-listed research questions.

4. Synopsis of New Framework

Effective for claims with dates of service on and after April 3, 2009, the CMS transitioned the prior framework—diagnosis, staging, restaging, and monitoring response to treatment—into the initial treatment strategy and subsequent treatment strategy framework. The chart below summarizes FDG PET coverage as of November 10, 2009:

FDG PET Coverage for Solid Tumors and Myeloma

Tumor Type	Initial Treatment Strategy (formerly "diagnosis" & "staging")	Subsequent Treatment Strategy (formerly "restaging" & "monitoring response to treatment")
Colorectal	Cover	Cover
Esophagus	Cover	Cover
Head & Neck (not thyroid or CNS)	Cover	Cover
Lymphoma	Cover	Cover
Non-small cell lung	Cover	Cover
Ovary	Cover	Cover
Brain	Cover	CED
Cervix	Cover w/exception*	Cover
Small cell lung	Cover	CED
Soft Tissue Sarcoma	Cover	CED
Pancreas	Cover	CED
Testes	Cover	CED
Breast (female and male)	Cover w/exception*	Cover
Melanoma	Cover w/exception*	Cover
Prostate	Non-Cover	CED
Thyroid	Cover	Cover w/exception or CED*
All other solid tumors	Cover	CED
Myeloma	Cover	Cover

All other cancers not listed

CED

CED

*Cervix: Non-covered for the initial diagnosis of cervical cancer related to initial treatment strategy. All other indications for initial treatment strategy are covered.

*Breast: Non-covered for initial diagnosis and/or staging of axillary lymph nodes. Covered for initial staging of metastatic disease. All other indications for initial treatment strategy are covered.

*Melanoma: Non-covered for initial staging of regional lymph nodes. All other indications for initial treatment strategy are covered.

*Thyroid: Covered for subsequent treatment strategy of recurrent or residual thyroid cancer of follicular cell origin previously treated by thyroidectomy and radioiodine ablation and have a serum thyroglobulin >10ng/ml and have a negative I-131 whole body scan. All other indications for subsequent treatment strategy are CED.

(This NCD last reviewed November 2009.)

Claims Processing Instructions

[TN 1879 \(Medicare Claims Processing\)](#) 

[TN 2873 \(Medicare Claims Processing\)](#) 

[TN 3911 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

110

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R110NCD.pdf> 

Revision History

09/2009 - CMS is adopting a coverage framework that replaces the four-part diagnosis, staging, restaging and monitoring response to treatment categories with a two-part framework that differentiates FDG PET imaging used to inform the initial antitumor treatment strategy from other uses related to guiding subsequent antitumor treatment strategies after the completion of initial treatment. CMS is making this change for all NCDs that address coverage of FDG PET for all oncologic conditions. Effective date 04/06/2009 Implementation date 10/19/2009 ([TN 106](#) ) (CR 6632)

10/2009 - This change request rescinds and replaces Transmittal 106, dated September 18, 2009. The effective date has been changed to April 3, 2009 and the implementation date has been changed to October 30, 2009. Business Requirements (BR) 6632.6.1 and 6632.6.2 have been revised to clarify that they are subsets of BR 6632.6 and are specific to CED. All other information remains the same.

12/2009 - Effective for claims with dates of service on and after November 10, 2009, CMS will end the coverage with evidence development requirements for FDG PET for cervical cancer and will cover only one FDG PET for cervical cancer for staging in beneficiaries with biopsy-proven tumors when the treating physician determines that the study is needed to determine the location and/or extent of the tumor for specific therapeutic purposes related to initial treatment strategy.

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Positron Emission Tomography \(FDG\) \(CAG-00065N\)](#) 
- [First reconsideration for Positron Emission Tomography \(NaF-18\) to Identify Bone Metastasis of Cancer \(CAG-00065R\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Alzheimer's Disease/Dementia \(CAG-00088N\)](#) 
- [First reconsideration for Positron Emission Tomography \(FDG\) and Other Neuroimaging Devices for Suspected Dementia \(CAG-00088R\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Breast Cancer \(CAG-00094N\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Thyroid Cancer \(CAG-00095N\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Myocardial Viability \(CAG-00098N\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Soft Tissue Sarcoma \(STS\) \(CAG-00099N\)](#) 
- [Original Consideration for Positron Emission Tomography \(N-13 Ammonia\) for Myocardial Perfusion \(CAG-00165N\)](#) 

- [Original Consideration for Positron Emission Tomography \(FDG\) for Brain, Cervical, Ovarian, Pancreatic, Small Cell Lung, and Testicular Cancers \(CAG-00181N\)](#)
- [First reconsideration for Positron Emission Tomography \(FDG\) for Solid Tumors \(CAG-00181R\)](#)
- [Second reconsideration for Positron Emission Tomography \(FDG\) for Cervical Cancer \(CAG-00181R2\)](#)
- [Third reconsideration for Positron Emission Tomography for Initial Treatment Strategy in Solid Tumors and Myeloma \(CAG-00181R3\)](#)
- [Original Consideration for Positron Emission Tomography \(FDG\) for Infection and Inflammation \(CAG-00382N\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Positron Emission Tomography (FDG) for Oncologic Conditions	4	06/11/2013 - N/A	View
Positron Emission Tomography (FDG) for Oncologic Conditions	3	08/04/2010 - 06/11/2013	View
Positron Emission Tomography (FDG) for Oncologic Conditions	2	11/10/2009 - 08/04/2010	You are here
Positron Emission Tomography (FDG) for Oncologic Conditions	1	04/03/2009 - 11/10/2009	View